



ESKISEHIR OSMANGAZI UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING
2021-2022 SPRING SEMESTER
DIGITAL SYSTEMS LABORATORY

EXPERIMENT 8
LOGIC DESIGN WITH GATE LEVEL AND DATA FLOW MODELING IN
VERILOG-HDL



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GROUP II – D

EXPERIMENT 8 REPORT

OBJECTIVES

- To get familiar with the gate-level modeling of combinational circuits in Verilog HDL
- To get familiar with the dataflow modeling of combinational circuits in Verilog HDL

INTRODUCTION

Dataflow modeling provides ways to describe combinatorial circuits by their function rather than their gate structure. Dataflow modeling uses a number of operators that act on operands to produce the desired results. Verilog HDL provides about 30 operator types. Gate level modeling is virtually the lowest level of abstraction because the switch-level abstraction is rarely used. Gate level modeling is used to implement the lowest-level modules in a design, such as multiplexers, full-adder, etc. Verilog has gate primitives for all basic gates.

PROCEDURE

1. Write gate-level Verilog HDL description of 2-to-4 line decoder with enable given in Figure 1 on paper with your handwriting. Specify the gates delays as 30, 20, and 10 ns for the gates AND, NOT and OR respectively. **Add your answer to your report.**

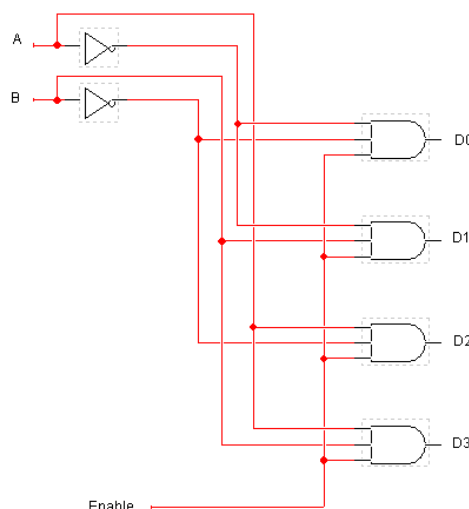


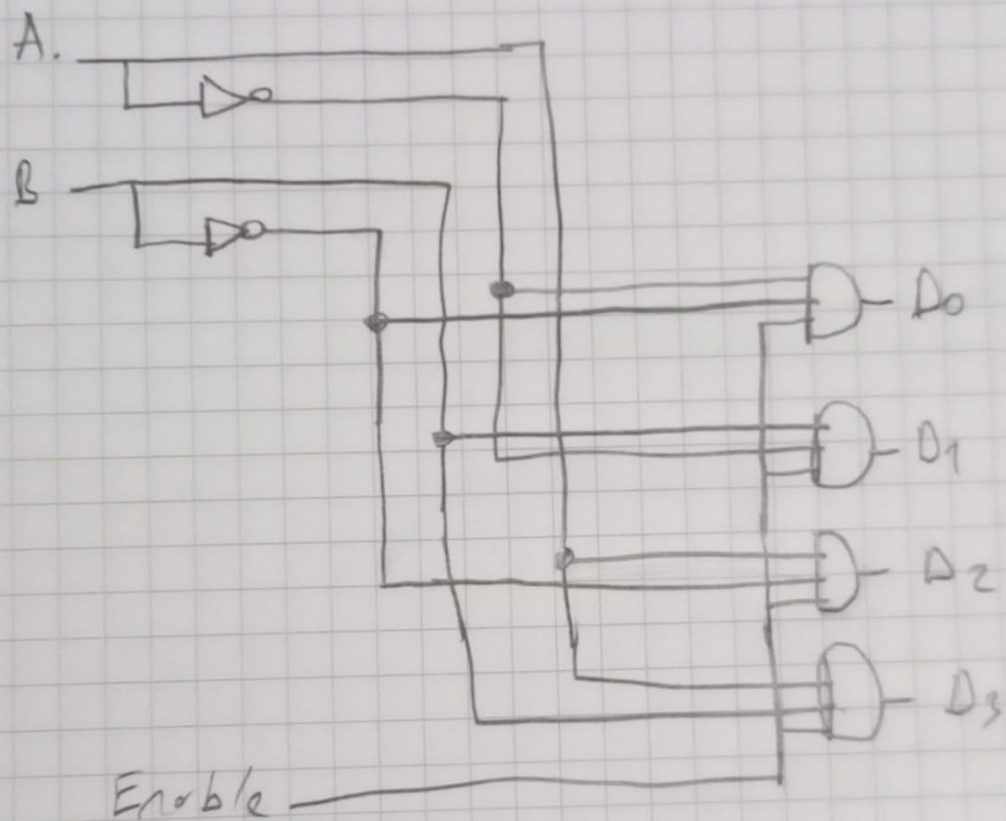
Figure 1: 2-to-4 line decoder with enable

Procedure

1-

Mustafa Koc

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```

Module Mustafa_Yogsci (A, B, EN, D0, D1, D2, D3);
input A, B, EN;
output D0, D1, D2, D3;

wire w1, w2;

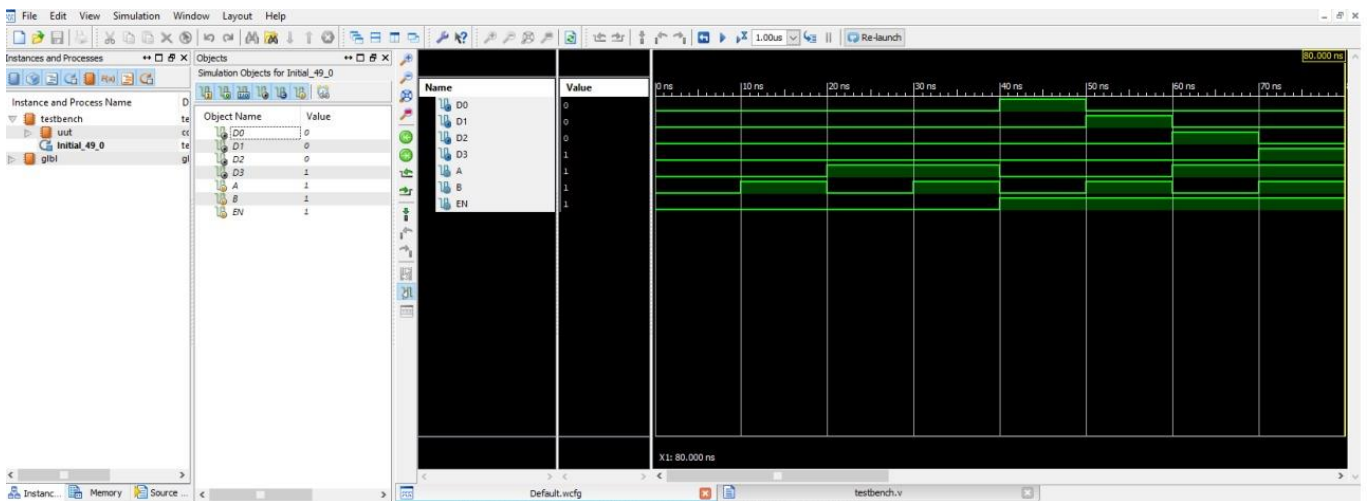
not # (20) g1 (w1, A);
not # (20) g2 (w2, B);
not # (20) g3 (D0, w1, w2, EN);
not # (20) g4 (D1, w1, B, EN);
not # (20) g5 (D2, A, w2, EN);
not # (20) g6 (D3, A, B, EN);

end module

```

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2. Enter gate-level Verilog HDL description of 2-to-4 line decoder that you write in Q1 in Xilinx ISE, and create a test bench file for it. All possible input combinations must be included in your test bench files. Inputs should be changed in every 10 ns. By using this test bench file, simulate your design and verify it. **Add your codes and simulation outputs to your reports.**



3. Write Verilog HDL dataflow description of a 2-to-1 line multiplexer with enable given in Figure 2 on paper with your handwriting, and enter it in Xilinx ISE. Create a test bench file for it. All possible input combinations must be included in your test bench files. Inputs should be changed in every 10 ns. By using this test bench, simulate your design and verify it. **Add your codes and simulation outputs to your reports.**

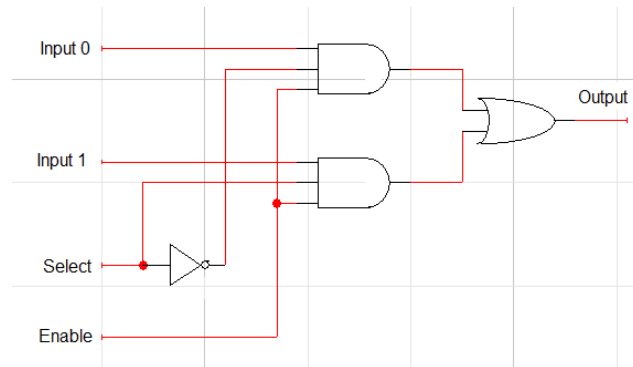


Figure 2: 2-to-1 line multiplexer with enable

Hint: You must add your name, surname and signature to the added papers including your handwritings. All projects will be named in format your NameSurname like AhmetDemir in your report.

```
module MustafaYogci (IO, I1, S, EN, OUT);  
  input IO, I1, S, EN;  
  output OUT;  
  assign OUT = IO & S & EN | I1 & ~S & EN;  
end module
```

Mustafa Yogci
15122013 2027
[Signature]

DISCUSSION AND CONCLUSIONS

As a result of the experiment, we tried to learn Gate-leveling and dataflow. We designed a 2-4 line decoder. We added the gate delays. But this with this delay we encountered problems that effected our output signals. In the third procedure we tried to 2-1-line multiplexer. But I couldn't this experiment